INCH-POUND

MIL-D-50677B (AR)
20 December 1990
SUPERSEDING
MIL-D-50677A (AR)
15 December 1979

MILITARY SPECIFICATION

DETECTOR, DOPPLER: 10548509

This specification is approved for use within the U. S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a doppler detector assembly used in Radar Receiver R-1475, part of Radar Set AN/VPS-2 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Main tenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - Standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.4)

U. S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

10548183 - Hybrid Junction
10548509 - Detector, Doppler

PACKAGING DATA SHEET

PS10548509 - Packaging of Doppler Detector

(Copies of drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

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3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 10548509, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the doppler detector shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.
b. Inorganic protective surface finishings.
c. Part identification and marking.
d. Workmanship.

3.3 Environmental service conditions.

3.3.1 Vibration. The doppler detector shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:

Frequency Range	Amplitude
10 to 15 hertz	0.06 inches double amplitude
15 to 25 hertz	0.04 inches double amplitude
25 to 55 hertz	0.02 inches double amplitude

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-55-10 hertz sweep time is 3 minutes 45 seconds ± 10%.

3.3.2 Shock. The doppler detector shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude.

3.3.3.1 Operating. The doppler detector shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.3.2 Non-operating. The doppler detector shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at 40,000 feet above sea level (5.5 inches of Hg).

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3.3.4 Temperature.

3.3.4.1 Operating. The doppler detector shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40 Degree F and +155 Degree F.

3.3.4.2 Non-operating (storage). The doppler detector shall not suffer permanent damage when subjected to the constant temperature extremes of -60 Degree F for a period of 24 hours and +160 Degree F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The doppler detector shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the doppler detector.

3.5 Performance. When supplied with the following signals of a through d and modified as required in accordance with 3.5.1 through 3.5.4, the performance of 3.5.1 through 3.5.4 will be obtained:

a. Connect 25.0 ± 0.1 volts direct current (Vdc) at 50 milliamperes (ma), with a shield to power supply case, to J1-1 (+), J1-2 (-) and J1-4 (shield) of A1 and A2 of the doppler detector assembly.

b. Supply signals to A5J1 and A6J1 which are equal in amplitude, from a source which is variable between 59.93660 megahertz (MHz) and 59.94660 MHz in frequency and from -120 to 0 dBm in amplitude, into a 50 ohm load.

c. Supply signals to A5J3 and A6J3 which are equal in amplitude, at a frequency of 59.93600 ± 0.00020 MHz, at a level of +2.5 ± 0.5 dBm into a 50 ohm load.

d. Load A1J1-A1 and A2J1-A1 with 150 ohms.

3.5.1 Noise. With the frequency of the signal applied to A5J1 and A6J1 set to 59.94000 ± .00020 MHz the signal measured at A1J1-A1 and A2J1-A1 shall be less than 31.0 millivolts (mV). The difference between the amplitudes measured at A1J1-A1 and A2J1-A1 shall not be more than 20 dB.

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3.5.2 Gain tracking.

a. When the signal level at A5J1 and A6J1 is increased to provide an output signal 5 dB above the greater noise level measured in 3.5.1, the difference in levels measured at A1J1-A1 and A2J1-A1 shall be less than 3 dB.

b. When the signal at A5J1 and A6J1 is increased to provide an output signal of 283.0 ± 17.0 mV at the greater output of a, the input signal shall have increased by a minimum of 20 dB. The level difference at A1J1-A1 and A2J1-A1 shall be less than 2 dB.

3.5.3 Signal tracking. When the signal level at A5J1 and A6J1 is set for -70 dBm, the phase difference between A1J1-A1 and A2J1-A1 shall be less than 10 degrees when the input frequency to A5J1 and A6J1 is varied from 59.93700 to 59.94500 MHz (doppler frequency of 1,000 to 9,000 Hz).

3.5.4 S.C.V.

a. An input signal shall be applied to A5J1 with a frequency of 59.93600 ± 0.00020 MHz, amplitude of -3.5 ± 0.5 dBm into 50 ohms, switched ON and OFF at a 20 KHz rate with the switching rise and fall times at 40.0 ± 5.0 nanoseconds (ns). The RMS output level at A1J1-A1 shall be noted.

b. When the frequency of the signal specified above is changed to 59.94000 ± 0.00020 MHz, the input level required to produce the same RMS output level as that noted in a, shall be at least 49.0 dB below the input level applied in a.

NOTE: The same requirement applies using A6J1 as the input and A2J1-A1 as the output.

3.6 Workmanship. The workmanship requirements of MIL-F-13926 shall be met except for soldering.

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement (task) F of paragraph 1.2 of MIL-STD-2000.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 °F to +90 °F
b.	Relative humidity	90% RH max.
c.	Pressure	28 to 32 inches of Hg

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4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three (3) Detector, Doppler for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table I (Functional tests) and Table II (Special sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and the same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examination and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies, and subassemblies shall be inspected in accordance with the inspection provisions contained in this specification and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table I shall be performed on a 100% basis.

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TABLE I. Functional tests (100 percent inspection).

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
101. Electrical	3.4	4.7.2
102. Performance	3.5	4.7.3
Minor:	None	N/A

4.5 Special sampling. One doppler detector out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table I and shall then meet the requirements and tests of Table II. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE II. Special sampling.

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
Critical:	None	N/A
Major:		
103. Vibration	3.3.1	4.7.1.1
104. Shock	3.3.2	4.7.1.2
105. Altitude	3.3.3	4.7.1.3
106. Temperature	3.3.4	4.7.1.4
107. Humidity	3.3.5	4.7.1.5
Minor:	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

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4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table III (Test equipment list) shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

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TABLE III. Test equipment list.

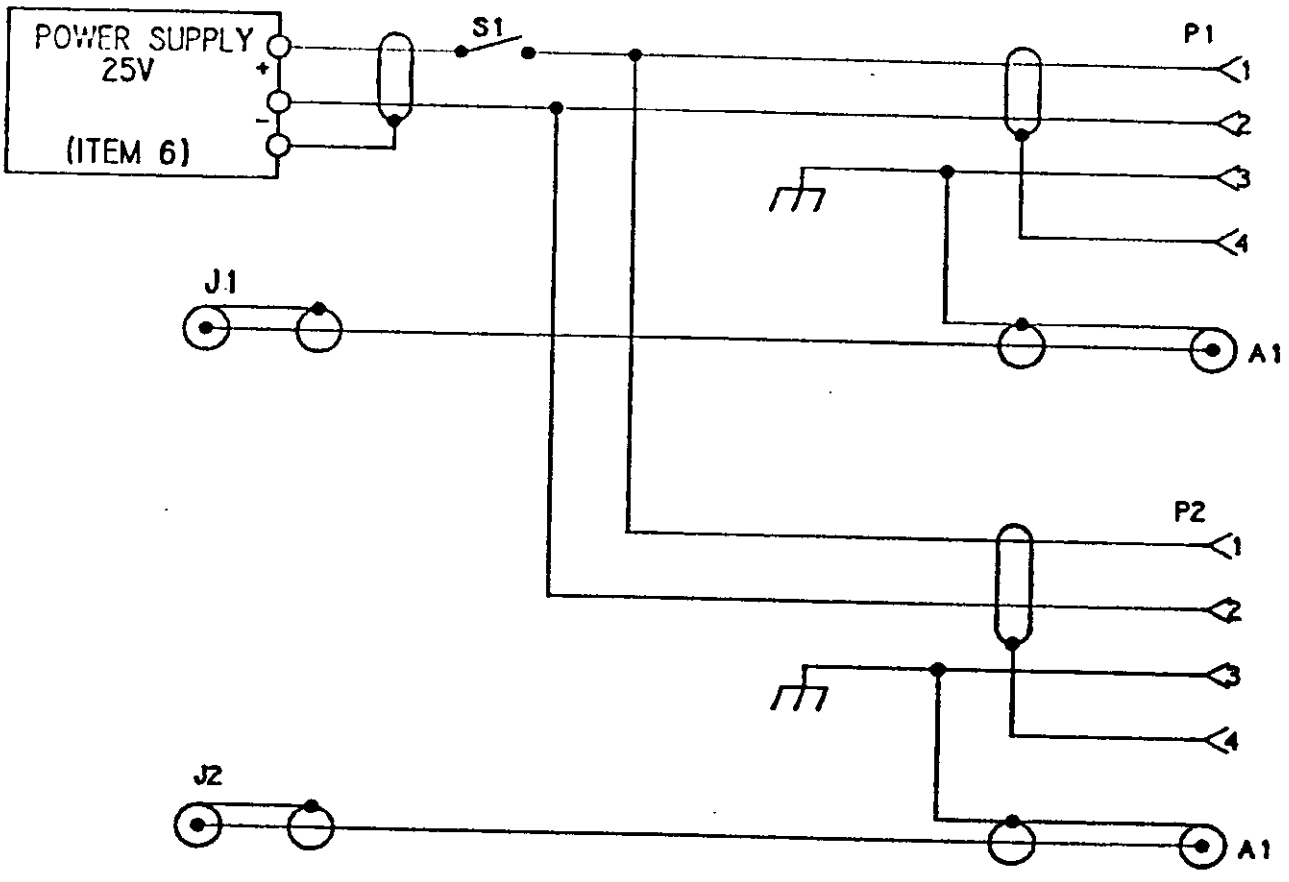
ITEM NUMBER	DESCRIPTION
1. RF Signal Generator (2)	Frequency: 60 MHz variable ±5 KHz Stability: 250 Hz after 15 min. warmup Output: To +3 dBm into 50 ohms
2. Phase Meter	Frequency: 600 Hz to 10 KHz min. Input signal range: 0.2 mV to 1 V RMS min. Display: ±18 Degree min. Resolution: 0.1 Degree Phase Accuracy: ±0.5 Degree Input Impedance: 1 Megohm min.
3. Frequency Counter	Frequency: 60 MHz Resolution: 25 Hz Sensitivity: 50 mV Input Impedance: 50 ohm
4. RMS Voltmeter	Range: 300 mV full scale Frequency: 10 Hz to 10 KHz minimum Input Impedance: 1 Megohm minimum Accuracy: ±1.0%
5. Variable Attenuator (2)	Range: 0 to 120 dB in 1 dB steps Impedance: 50 ohms Accuracy: ±0.5 dB of setting
6. Power Supply	Output: 24-26 volts at 50 ma Ripple: Less than 5 mV
7. Oscilloscope	Frequency Response: 1 MHz Time Base: 1 us to 100 ms/cm Sensitivity: 20 mV to 1 V/cm Vertical Amplifier: Dual channel Modes: Chan 1, chan 2, copped, add, AC/DC coupled
8. Digital DC Voltmeter	Range: 10-30 volts Accuracy: ±0.1%
9. R.F. Voltmeter	Range: 300 mV full scale Frequency: 40 to 80 MHz Accuracy: ±2% Input Impedance: 50 ohms

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TABLE III. Test equipment list (continued).

ITEM NUMBER		DESCRIPTION
10.	Pulse Generator	Repetition Note: 20.0 ± 2.0 KHz Pulse Width: 0.4 ± 0.02 us Amplitude: +0.50 ± 0.02 V Rise and Fall Time: 40.0 ± 5.0 ns
11.	Test Harness	See Figure 1
12.	Hybrid Junction	Frequency: 60 MHz Outputs: Two (2) Coupling: -3 dB Isolation: 30 dB min. Amplitude Balance: 0.2 dB max. Insertion Loss: 0.75 dB max. Impedance: 50 ohms VSWR: 1.3:1 max. P/N 10548183
13.	Termination (2)	150 ohm
14.	BNC Tee (2)	UG-274/U
15.	Power Divider	Frequency: 60 MHz Outputs: Four (4) Coupling: -6 dB Isolation: 30 dB min. Amplitude Balance: 0.2 dB max. Insertion Loss: 0.7 dB Impedance: 50 ohms VSWR: 1.3:1 max.
16.	Termination (2)	50 ohm
17.	Double Balanced Mixer	Frequency: LO 60 MHz RF 60 MHz Impedance: 50 ohms
18.	Phase Shifter	Frequency: 60 MHz Bandwidth: 10% Phase Shift Range: 0-180 min. VSWR: 1.5:1 max. Impedance: 50 ohms Insertion Loss: 1 dB max.

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NOTES:

1. P1 AND P2 = ITT CANNON P/N DEM-5W1S.
2. J1 AND J2 = BNC MALE, UG-88/U OR EQUIVALENT.
3. S1 = CUTLER-HAMMER 7580K7 OR EQUIVALENT.

FIGURE 1. Test harness.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The doppler detector shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the doppler detector shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10% amplitude and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 Hertz.

After the application of all the shock impulses, the doppler detector shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating and non-operating). This test shall be performed as specified in Procedures II and III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

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4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The doppler detector shall be tested as follows:

a. Connect the equipment as shown in Figure 2. Set PWR switch on test harness (Table III, item 11) to OFF. The test equipment shall have a 15 minute warmup. The two RF signal generators (Table III, item 1) are monitored with frequency counters (Table III, item 3) until the signals stabilize to allow a test to be completed with less than 250 Hz total relative drift.

b. Set Attenuator #1 (Table III, item 5) to 3 dB. Connect RF voltmeter (Table III, item 9) to J4 of the Power Divider (Table III, item 15) and adjust Signal Generator #1 (Table III, item 1) for 300.0 $\pm$ 5.0 mV read on the RF Voltmeter (Table III, item 9) at a frequency of 59.93600 $\pm$ 0.00020 MHz as read on the frequency counter.

c. Set Signal Generator #2 (Table III, item 1) output level for -17 dBm as indicated on the signal generator meter, and a frequency of 59.9400 $\pm$ 0.00020 MHz as indicated on the frequency counter. Set Attenuator #2 (Table III, item 5) to 101 dB.

d. Set Power Supply (Table III, item 6) for 25.0 $\pm$ 0.1 Vdc as measured with the digital voltmeter (Table III, item 8).

4.7.3.1 Noise. With the equipment set up as in 4.7.3a, b, c and d, turn PWR switch on the test harness (Table III, item 11) to ON. Connect J1 (loaded with 150 ohms, Table III, item 13) to the RMS Voltmeter (Table III, item 4) and record the output of amplifier A1. LIMIT: 31.0 mV maximum. Disconnect J1 and connect J2 to the RMS Voltmeter (loaded with 150 ohms) and record the output of the amplifier A2. LIMIT: 31.0 mV maximum. Calculate the difference between the two readings. LIMIT: 3.0 dB maximum.

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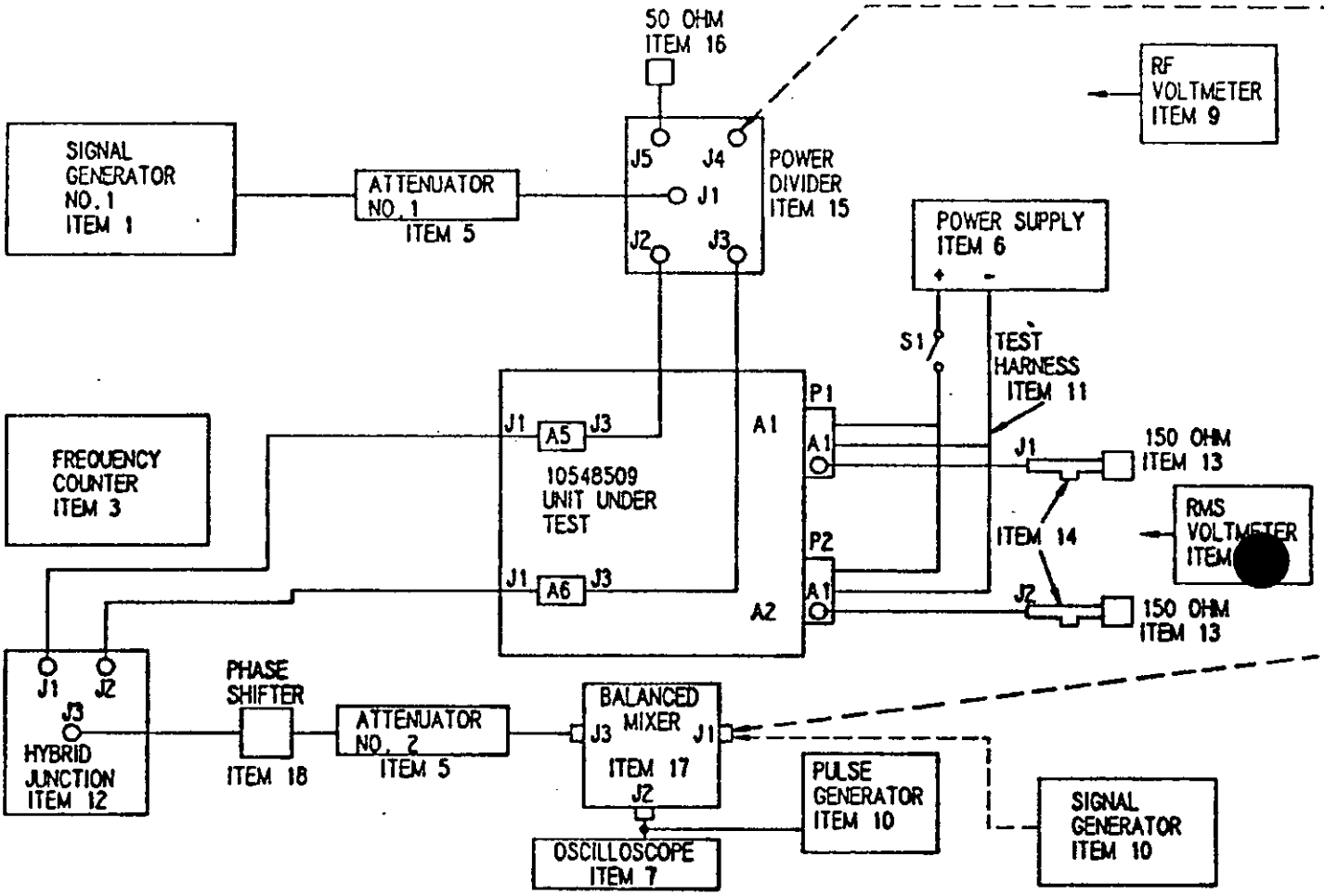


FIGURE 2. Doppler detector test setup.

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4.7.3.2 Gain tracking.

a. Connect the equipment so the larger noise level of 4.7.3.1 is being monitored on the RMS voltmeter. Reduce attenuator #2 (Table III, item 5) for an output reading 5 dB higher than the noise level measured in 4.7.3.1. Record the setting of attenuator #2. Record the level at both J1 and J2 and calculate the difference. LIMIT: 3.0 dB maximum.

b. Connect the larger signal of J1 and J2 to the RMS voltmeter and reduce attenuator #2 for an output reading of 283.0 ± 17.0 mV. Record the setting of attenuator #2. It shall be at least 20 dB less than the setting of 4.7.3.2. Record the output level at both J1 and J2 and calculate the difference. LIMIT: 2.0 dB maximum.

4.7.3.3 Signal tracking.

a. Connect the equipment as shown in Figure 2. Set signal generator #1 and attenuator #1 as specified in 4.7.3.b. Set signal generator #2 frequency for 59.93660 ± 0.00020 MHz with an output level of -10 dBm. Set attenuator #2 for 60 dB of attenuation. Connect J1 and J2 (loaded with 150 ohms) to the A and B inputs of the phase meter (Table III, item 2).

b. Vary signal generator #2 to each of the frequencies listed below and record the phase angle reading at each frequency. Subtract the lowest phase angle reading from the highest. The difference shall not exceed 10 degrees.

FREQUENCY (+0.00020 MHz)

59.94100 MHz	
59.93700 MHz	59.94200 MHz
59.93800 MHz	59.94300 MHz
59.93900 MHz	59.94400 MHz
59.94000 MHz	59.94500 MHz

4.7.3.4 S.C.V. Connect equipment as shown in Figure 3. Set the pulse generator (Table III, item 10) as follows:

Repetition Rate	- 20.0 ± 2.0 KHz
Pulse Width	- 0.40 ± 0.02 us
Amplitude	- +0.50 ± 0.02 V
Rise Time	- -40.0 ± 5.0 ns
Fall Time	- -40.0 ± 5.0 ns

Set signal generator #1 and attenuator #1 as specified in 4.7.3.b. Set signal generator #2 to 59.94000 ± 0.00020 MHz with an output level of 300.0 ± 5.0 mV as measured with the RF Voltmeter (Table III, item 9).

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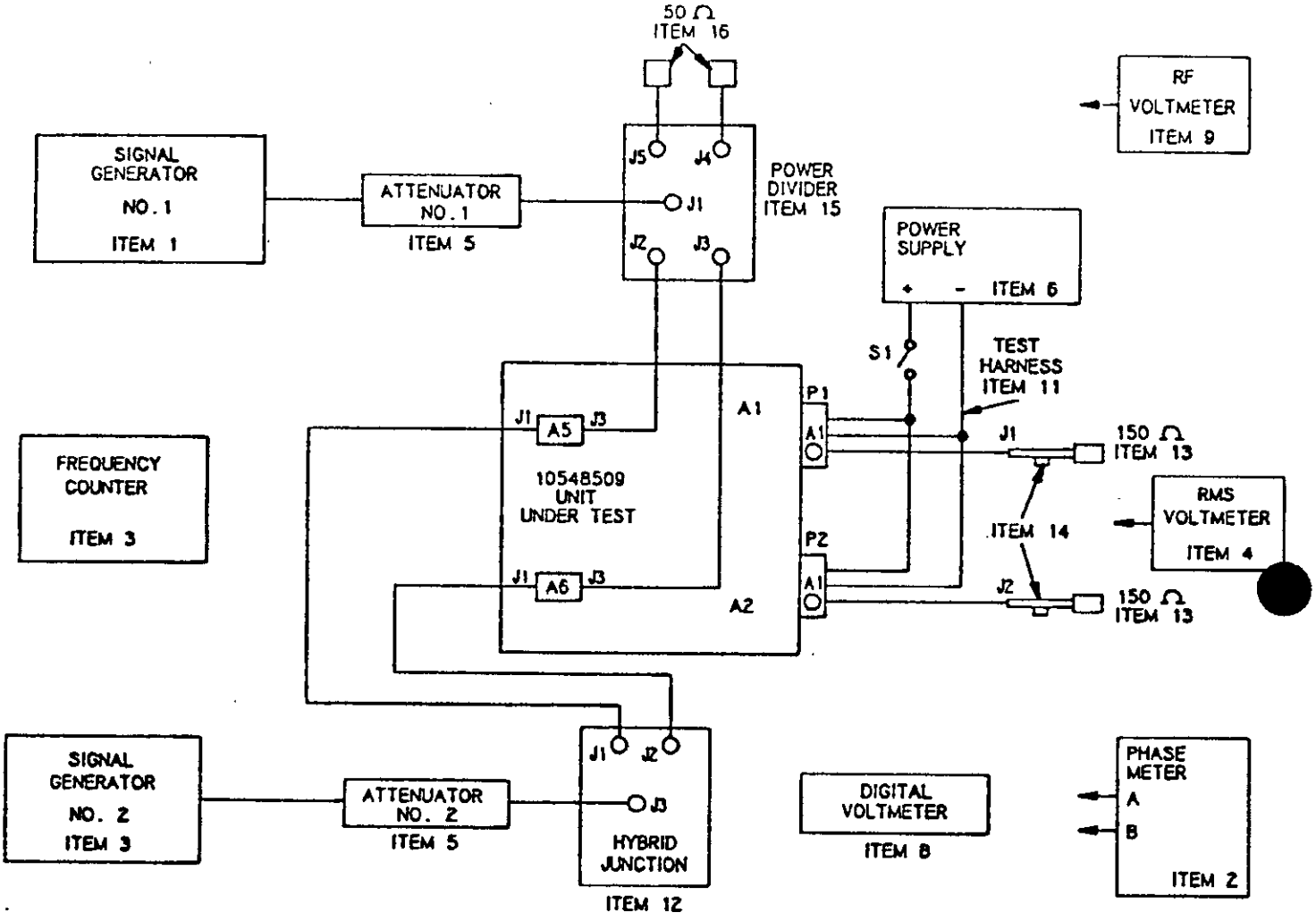


FIGURE 3. Test setup, SCV.

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4.7.3.4.1 Connect J4 of power divider (Table III, item 15) to J1 of the balanced mixer (Table III, item 17). Adjust attenuator #2 for 0 dB attenuation. Adjust the phase shifter (Table III, item 18) for a peak J1 indication on the RMS voltmeter (Table III, item 4). Record the RMS voltmeter reading.

4.7.3.4.2 Set attenuator #2 for 60 dB of attenuation. Disconnect the power divider J4 from J1 of the balanced mixer and connect the signal generator #2 output to J1 of the balanced mixer. Adjust attenuator #2 for the same RMS voltmeter reading of 4.7.3.4.1 Record the attenuator #2 total attenuation. LIMIT: 49.0 dB minimum.

4.7.3.4.3 Connect the equipment so J2 output is being monitored on the RMS voltmeter. Repeat steps 4.7.3.4.1 and 4.7.3.4.2 for A2.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of MIL-STD-2000 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Packaging Data Sheet PS10548509.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The doppler detector is used to remove any doppler effect from the i-f frequency of radar return signals in Radar Set AN/VPS-2. This doppler is caused by the transmitted rf frequency of the radar striking a moving target and the return frequency being shifted in frequency. This frequency shift is used to obtain an indication of moving targets.

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6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

a. Title, number and date of this specification.
b. Applicable stock number.
c. First article sample requirements.
d. Packaging requirements, if other than specified in Section 5.
e. Serialization requirements, if applicable.

6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, Attn: AMSMC-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (Key word) listing.

Fire Control System
Radar Receiver R-1475/VPS-2
Radar Set AN/VPS-2
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A676)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-D-50677B (AR)	2. DOCUMENT DATE (YYMMDD) 20 December 1990

3. DOCUMENT TITLE
DETECTOR, DOPPLER: 10548509

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army ARDEC Standardization & Specification Office	b. TELEPHONE (Include Area Code) (1) Commercial: (201) 724-6674 (2) AUTOVON: 880-6674
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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